

Operating System – Sessional Exam Analysis

Past-paper analysis mapped to your sessional syllabus | Papers analysed: Jan 2022, Feb 2022 (Supplementary), Jan 2023, May 2024, Jan 2025 | Nirma University, MCA Semester I

66%

of past-paper marks (265 of 400) fall inside your sessional syllabus. The rest is scheduling, memory management, disk and file topics, which are not in your deck – skip them.

5 / 5

papers contained a Deadlock question; 4 of 5 contained a Threads question and a semaphore pseudo-code question.

75%

of in-scope marks come from just 5 topics: Deadlock, Semaphores / Sync, Threads, Process model, and OS fundamentals + kernel (198 of 265 marks).

Legend: J22 = Jan 2022, F22 = Feb 2022 Supplementary, J23 = Jan 2023, M24 = May 2024, J25 = Jan 2025. Q-numbers follow each paper (e.g. J23 Q4C = Jan 2023, question 4, part C). Slide numbers are page numbers of OS_Sessional_Syllabus.pdf.

1. Scope check – what in the past papers is actually in your sessional?

Your syllabus deck covers six chapters: Computer System Overview, OS Overview, Process Description & Control, Threads/SMP/Microkernels, Concurrency (Mutual Exclusion & Synchronization) and Deadlock & Starvation. I filtered every past-paper question against it (58 question parts in total, counting OR alternatives).

Paper	Total marks	In-scope marks	% in scope	Questions OUTSIDE the sessional syllabus
Jan 2022 (J22) – 3MCA103, 2 hrs	50	43	86%	Q5 page replacement / fixed-vs-dynamic partitioning (7). Q7 option B (SPN & SRT scheduling) is out; Q7 option A (DMA) is IN, so take that choice.
Feb 2022 Supplementary (F22) – 3MCA103, 2 hrs	50	35	70%	Q5 simple paging numerical / dynamic partitioning, best-fit & next-fit (7). Q7 Round-Robin & SPN, or SSTF & SCAN disk scheduling (8) – both options out.
Jan 2023 (J23) – 6CS103, 3 hrs	100	73	73%	Q4B first-fit vs best-fit (7); Q5A logical/relative/physical address (7); Q5B rotational latency, FIFO & SSTF disk (7); Q6A pre-emptive vs non-pre-emptive, SPN (6).
May 2024 (M24) – 6CS103, 3 hrs	100	58	58%	Q4B internal vs external fragmentation (8); Q5A FIFO/LRU page replacement (8); Q5B file allocation (8); Q6A FCFS/SSTF disk (8); Q6B SPN & Round-Robin Gantt charts (10).
Jan 2025 (J25) – 6CS103CC22, 3 hrs	100	56	56%	Q1B types of schedulers (8); Q2B FCFS vs Round-Robin (6); Q3B Optimal/LRU page replacement (8); Q4B paging & MMU (8); Q5C page-fault handling (6); Q6A FCFS/SSTF disk (8).
TOTAL	400	265	66%	Matches your "roughly 60%" estimate for the sessional portion.

Frequently asked, but NOT in your sessional – do not spend time here

Topic	Appeared in	Why it is excluded
CPU scheduling (FCFS, Round-Robin, SPN, SRT, pre-emptive vs non-pre-emptive, types of schedulers)	All 5 papers	Separate scheduling chapter – not in your deck.

Topic	Appeared in	Why it is excluded
Memory management: partitioning, fragmentation, paging & address translation, page replacement, page faults	J22, F22, J23, M24, J25	Only a conceptual 1-slide mention of paging (slide 116) – no algorithms or numericals.
Disk scheduling (FCFS, SSTF, SCAN, rotational latency)	F22, J23, M24, J25	I/O & disk management chapter – not in your deck.
File allocation methods	M24	File-management chapter – not in your deck.

2. Topic weightage table

Frequency = number of the 5 papers in which the topic appeared (counted once per paper). Typical marks = marks assigned to that question part in the papers. Row colour: green = 3+ papers, amber = 2, light yellow = 1, grey = never asked.

Unit / Chapter	Specific topic	Freq. (of 5)	Where it appeared	Typical marks	Deck slides
Ch 1 – Computer System Overview	Multiprocessor vs multicore; SMP architecture + diagram; advantages of SMP over uniprocessor	3 / 5	J22 Q6 · F22 Q6 · J23 Q1C	6 – 7	68–74, 132, 251–253
	Interrupts: purpose, interrupt cycle, types, simple vs multiple interrupts	2 / 5	M24 Q1A · J25 Q5B	6 – 8	26–42
	Instruction-format numerical: addressable memory, PC & IR width, bus-width effect on speed	2 / 5	J23 Q6B · J25 Q6B	4 – 8	17–25, 75–77
	I/O techniques (programmed, interrupt-driven, DMA); DMA block diagram & configurations	1 / 5	J22 Q7	8	64–67
	Cache memory: structure, read operation	1 / 5	J25 Q2C	6	53–63
	Top-level computer components, registers, block diagram	1 / 5	J23 Q1A	7	8–14
	Memory hierarchy, locality, access-time & cost numericals (Practice Exercise 2)	0 / 5	Never asked	–	44–52, 78
Ch 2 – Operating System Overview	OS definition, objectives, services, resources managed, examples, types / evolution	3 / 5	J22 Q1 · F22 Q1 · J25 Q1A	7 – 8	80–103
	OS storage-management responsibilities; virtual memory overview	1 / 5	M24 Q1B	8	113–118
	Modern OS design, Windows / UNIX / Linux architecture	0 / 5	Never asked	–	130–150
Ch 3 ("Unit-2") – Process Description & Control	Process vs program; 5-state & 7-state model with diagram; transitions; reasons for suspension	3 / 5	F22 Q2 · M24 Q2A · J25 Q2A	6 – 8	155–177
	Process image; PCB elements	1 / 5	J23 Q2A	7	183–196

Unit / Chapter	Specific topic	Freq. (of 5)	Where it appeared	Typical marks	Deck slides
	Process creation / termination, mode & process switch, OS control tables, UNIX process states	0 / 5	Never asked	–	165–167, 178–182, 197–210
Ch 4 – Threads, SMP & Microkernels	Thread concept; thread vs process; single- vs multi-threaded model; multithreading vs multiprocessing	3 / 5	J22 Q3 · J23 Q2B · M24 Q2B	7 – 8	212–222
	User-level (ULT) vs kernel-level (KLT) threads: advantages, disadvantages, comparison	3 / 5	J22 Q3 · J23 Q2C · J25 Q5A	6 – 7	227–239
	Kernel; monolithic vs microkernel + one example OS of each	2 / 5	F22 Q1 · J23 Q1B	7	88, 130, 254–260
Ch 5 – Concurrency: Mutual Exclusion & Synchronization	Semaphores (binary vs counting) + producer/consumer pseudo-code (bounded, infinite, circular buffer)	3 / 5	F22 Q3 · J23 Q4C · J25 Q4A	6 – 8	304–325
	Process interaction (degrees of awareness); mutual exclusion; critical section	2 / 5	J22 Q2 · J23 Q3	7 – 10	280–293
	Readers/Writers problem with semaphores (writers' priority)	1 / 5	M24 Q3A	8	343–345
	Hardware support for mutual exclusion, monitors, message passing	0 / 5	Never asked	–	294–302, 326–342
Ch 6 – Concurrency: Deadlock & Starvation	Deadlock: definition, necessary conditions (+ circular wait), deadlock vs livelock	4 / 5	J22 Q4 · F22 Q4 · J23 Q4A · M24 Q4A	7 – 8	348–364
	Banker's algorithm: need matrix, safe state / safe sequence, resource-request check	2 / 5	J23 Q5C · M24 Q3B	6 – 10	368–382
	Resource allocation graph / deadlock modelling; consumable vs reusable; recovery	2 / 5	J22 Q4 · J25 Q3A	7 – 8	353–359, 389–390
	Prevention vs avoidance vs detection; deadlock prevention in detail	1 / 5	F22 Q4	7	365–391
	Deadlock-detection algorithm; dining philosophers	0 / 5	Never asked	–	384–388, 392–397

A question that covers two topics (e.g. J22 Q3 asks about threads and ULT vs KLT) is counted once for each topic. Where a paper offered an OR choice, the in-scope option is counted.

Where the marks actually are (in-scope marks only: 265)

Rank	Topic family	Marks (of 265)	Share	Papers	Note
1	Deadlock (conditions, prevention, Banker's, RAG, recovery)	53	20.0%	5 / 5	Q4 in 4 papers; Banker's numerical in J23 & M24; RAG in J22 & J25
2	Hardware basics (interrupts, I/O & DMA, cache, instruction numericals, components)	47	17.7%	4 / 5	Heavier in the recent papers (M24, J25)

Rank	Topic family	Marks (of 265)	Share	Papers	Note
3	Concurrency & synchronization (semaphores, producer/consumer, readers/writers, mutual exclusion)	46	17.4%	5 / 5	Pseudo-code question in F22, J23, M24, J25
4	OS fundamentals (definition, objectives, types, kernel, storage management)	37	14.0%	5 / 5	A definitional question appears in every paper
5	Threads (thread concept, ULT vs KLT)	34	12.8%	4 / 5	Two sub-questions in J22 and J23
6	Process model (states, suspension, process image / PCB)	28	10.6%	4 / 5	Absent only from J22
7	Multiprocessor / multicore / SMP	20	7.5%	3 / 5	J22, F22, J23 – not in M24 or J25

Jan 2022 Q2 (process + mutual exclusion + critical section + turnaround time) is counted under Concurrency; turnaround time itself is a scheduling term and is not in your deck.

3. High-yield priority list – study these first

Ranked by number of papers, total marks and how predictable the question format is. Together these five topics account for about 75% of in-scope marks.

Rank	Topic	Why it is high-yield	What to master (exam checklist)	Time share
1	Deadlock	Appears in 5 of 5 papers; 53 marks (20%) of in-scope marks. Mix of theory (conditions), numerical (Banker's) and diagram (RAG) – predictable and scoring.	- Definition + the 3 necessary conditions (mutual exclusion, hold-and-wait, no pre-emption) + the 4th condition, circular wait – the exam literally asks "three + fourth" (J23). - Prevention: how to break each condition (all resources at once; release & re-request; resource ordering) with the drawbacks (slides 365–367). - Prevention vs avoidance vs detection – prepare a 3-column comparison table (F22). - Banker's algorithm: $\text{Need} = \text{Max} - \text{Allocation}$; find safe sequence step by step; for a new request, check $\text{Request} \leq \text{Need}$ and $\leq \text{Available}$, tentatively allocate, re-run safety test (exact M24 template). - Resource allocation graph (draw + spot cycle), reusable vs consumable resources, 4 recovery strategies (slide 389), deadlock vs livelock.	25%
2	Semaphores & classical synchronization problems	4 of 5 papers had a semaphore pseudo-code question; 46 marks (17%) incl. mutual exclusion theory. Each paper rotates the variant, so learn all of them.	- semWait / semSignal primitives; binary vs counting semaphore; mutual exclusion using a semaphore (slides 304–309). - Producer/consumer in all 3 variants: infinite buffer with binary semaphores (J23), bounded buffer with counting semaphores (F22, J25), circular buffer (slides 322–325 – likeliest next variant). - Readers/Writers with writers' priority (M24) and be ready for the readers'-priority twist. - Process interaction: 3 degrees of awareness, with relationship, influence and potential problem for each (J23, 10 marks). - Be able to explain WHY each semaphore exists and what goes wrong if the order of semWait calls is swapped.	22%

Rank	Topic	Why it is high-yield	What to master (exam checklist)	Time share
3	Threads	4 of 5 papers; 34 marks (13%) . ULT vs KLT has been asked 3 times almost verbatim.	- Thread definition: unit of dispatching vs process as unit of resource ownership. - Diagram: single-threaded vs multi-threaded process model (J23). - Thread vs process table; multithreading vs multiprocessing (M24). - ULT vs KLT: advantages (no mode switch, app-specific scheduling, runs on any OS) vs disadvantages (one blocking system call blocks all threads; no multi-processor use); KLT is the reverse; combined approach (Solaris).	15%
4	Process model: states, suspension, process image	4 of 5 papers; 28 marks (11%) . Always diagram-based.	- Process vs program; 5-state diagram with every transition labelled (admit, dispatch, time-out, event wait, event occurs, release). 7-state model: add Blocked/Suspend and Ready/Suspend with transitions (M24); reasons for suspension (J25). - Process image = program + data + stack + PCB; PCB's 3 groups: identification, processor-state information, process-control information (J23).	12%
5	OS fundamentals & kernel / microkernel	5 of 5 papers open with a definitional question; 37 marks (14%) . Easy marks if you have crisp definitions.	- OS definition, objectives (convenience, efficiency, ability to evolve), services, resources managed, 2 example OS names (J22, F22, J25). - Types / evolution: serial → simple batch → multiprogrammed batch → time-sharing (slides 89–103). - Kernel; monolithic vs microkernel comparison with an example OS of each (F22, J23). Note: "monolithic" is not on any slide – use your notes / textbook. - Five storage-management responsibilities: process isolation, automatic allocation, modular programming, protection & access control, long-term storage (M24).	10%

Time share = suggested portion of your total revision time. The remaining 16% is for the quick wins below.

Quick wins – smaller topics worth a short revision slot (suggested 16% of time)

Topic	30-second cheat sheet
SMP & multicore (3/5, 20 marks)	Draw SMP organisation (processors with caches → shared bus → memory & I/O). SMP traits: 2+ similar processors, shared memory & I/O, equal access time, one integrated OS. Advantages: performance, availability, incremental growth, scaling. Not asked in M24 / J25 – do not skip.
Interrupts (2/5 – the two latest papers)	Interrupt cycle; save context → run handler → restore. Simple interrupt processing vs multiple interrupts: sequential (disable interrupts) vs nested (priority-based). Why needed: processor is faster than I/O.
Instruction-format numerical (2/5 – same question in J23 & J25)	Deck solution (slides 75–77): address bits = 32 – 8 (opcode) = 24 → $2^{24} = 16 \text{ MB}$; PC = 24 bits; IR = 32 bits; 16-bit data bus needs 2 bus cycles per 32-bit fetch; a 16-bit address bus adds a two-step address transfer.
DMA & cache (1/5 each)	DMA: I/O module transfers a block directly to/from memory, interrupts the processor only at the end. Cache read operation: processor gives address → hit: word delivered; miss: block fetched from main memory into cache, then word delivered.

4. Recurring question patterns

#	Pattern	Evidence from the past papers	How to use it
1	The opening question is definitional and multi-part	J22 Q1 and F22 Q1 are "What is an OS...?" + list/compare; J23 Q1A–C and M24 Q1A–B are 6–8 mark definitional parts; J25 Q1A "What is an OS? Explain types."	Memorise tight definitions with 3–5 supporting bullet points. This is the cheapest section to score full marks in.
2	"Differentiate / compare" is the single most common format	ULT vs KLT (3×), multiprocessor vs multicore (3×), monolithic vs microkernel (2×), process vs program, thread vs process, binary vs counting semaphore, consumable vs reusable, deadlock vs livelock, prevention vs avoidance vs detection.	Keep one 2-column comparison table per pair; examiners reward point-for-point contrast.
3	"Draw a neat diagram" is attached to almost every theory question	5-state / 7-state diagram; SMP organisation; single vs multi-threaded process; deadlock model / RAG; DMA block diagram; cache read operation; process image.	Practise redrawing these 7 diagrams from memory – a labelled diagram is usually 2–3 of the 7 marks.
4	Semaphore pseudo-code with a rotating variant	F22: bounded buffer, counting semaphores · J23: infinite buffer, binary semaphores · M24: readers/writers, writers' priority · J25: finite buffer.	Learn the family of solutions, not one program. Practise a quick change: swap buffer type, semaphore type or priority rule.
5	Two stock numericals recur	Banker's / safe state: J23 Q5C, M24 Q3B (10 marks). 32-bit microprocessor PC/IR: J23 Q6B, J25 Q6B (extended with bus-width and max-memory parts).	Solve the M24 Banker's snapshot and slide 75 exercise end-to-end until they are mechanical.
6	Questions are repeated almost verbatim across years	SMP question text is near-identical in J22 Q6, F22 Q6 and J23 Q1C; ULT/KLT in J22, J23, J25; deadlock conditions in four papers; 32-bit microprocessor in J23 and J25.	Use past-paper wording as your question bank – the sessional will likely recycle it.
7	Each 6–8 mark question has 2–4 hidden sub-asks	"What is a thread? Explain ULT and KLT discussing advantages and disadvantages" (J22 Q3); "Define deadlock... conditions... fourth condition" (J23 Q4A); "What is an interrupt? How are simple and multiple interrupts dealt with?" (M24 Q1A).	Answer in labelled sub-parts that mirror the wording; each sub-ask carries marks.
8	Question text follows the slide titles	"Process Image", "Reasons for Process Suspension", "Advantages / Disadvantages of ULTs", "Readers/Writers Problem", "Banker's algorithm", "Process Interaction" are all slide headings.	Treat slide titles as the question list – study heading by heading. Exception: "monolithic kernel" is asked but not on a slide.
9	Higher Bloom's levels sit on deadlock and synchronization	Deadlock / semaphore / Banker's questions are tagged BL3–BL5 (apply / analyse); definitions and architecture are BL1–BL2.	For these topics, practise worked solutions and tracing, not only reading theory.
10	Recent papers add "twists"	M24: writers' priority, deadlock vs livelock, Banker's request check. J25: consumable vs reusable resources, RAG scenario with unit counts, deadlock recovery.	Go one step beyond textbook basics: variants, edge cases and recovery.

5. Likely sessional questions (practice list)

Built from frequency, recency and the patterns above. Treat this as a practice list, not a guarantee.

Likelihood	Practise answering...	Basis
Very high	Define deadlock. State the necessary conditions. Explain deadlock prevention (how each condition is broken).	Conditions asked in 4/5 papers; prevention in detail in F22.
Very high	Banker's algorithm: compute the Need matrix, decide whether the state is safe (give the safe sequence), and test whether a new request can be granted.	J23 Q5C and M24 Q3B; numericals are favoured in later papers.
Very high	Write semaphore pseudo-code for the producer/consumer problem (bounded / infinite / circular buffer) and explain it.	F22, J23, J25 – a new variant each time.
Very high	Compare user-level and kernel-level threads with advantages and disadvantages.	J22, J23, J25.
High	What is a thread? How does it differ from a process? Single- vs multi-threaded model; multithreading vs multiprocessing.	J22, J23, M24.
High	Draw the 5-state / 7-state process model and explain transitions and reasons for process suspension.	F22, M24, J25.
High	Define an OS, its objectives and types; explain kernel and monolithic vs microkernel with an example.	A definitional question in all 5 papers.
Medium–High	Multiprocessor vs multicore; SMP architecture with diagram; advantages over uniprocessor.	J22, F22, J23 – missing from the two latest papers, so do not skip.
Medium	Interrupts: why required, types, simple vs multiple interrupt handling.	M24 Q1A, J25 Q5B.
Medium	Process image and PCB: elements and their roles.	J23 Q2A (7 marks).
Medium	Readers/Writers solution using semaphores – readers' priority and writers' priority.	M24 asked writers' priority; the other priority is the natural variant.
Medium	32-bit microprocessor numerical: max addressable memory, bus-width impact, PC and IR bits.	J23 and J25 – same question, extended.
Wildcard	Dining philosophers; monitors; message passing; compare-and-swap / exchange; deadlock-detection algorithm; memory-hierarchy and cache numericals (slides 48–52, 78).	Never asked in 5 papers, but the deck has dedicated slides (dining philosophers has 5 slides + an exercise) – a sessional setter could draw from them.

6. Assumptions and caveats

- The past papers are semester-end exams (50 or 100 marks). Your sessional may use a different mark scheme, so use the frequencies to prioritise topics rather than to predict exact marks.
- Two J25 questions are treated as outside your syllabus because the deck only mentions them in passing: types of schedulers (Q1B) and paging in the MMU (Q4B). If your faculty covered them in class, add them to your list.
- "Monolithic kernel" has been asked twice (F22 Q1, J23 Q1B) but is not on any slide of the deck. Use your class notes or the textbook for that comparison.
- Some slides are diagram-only (5-state / 7-state models, resource allocation graphs, reasons for suspension on slide 176), so I could only read their titles. Review them visually; the usual textbook (Stallings) suspension reasons are swapping, other OS reason, interactive user request, timing and parent process request.